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COMPRESSOR FOR CO2 CYCLE WITH AT LEAST TWO CASCADE COMPRESSION STAGES FOR ASSURING SUPERCRITICAL CONDITIONS

Abstract

The compressor is used for processing a CO2 flow; a first compressor stage has a first row of blades with a first number of blades and a second compressor stage, downstream the first compressor stage, has a second row of blades with a second number of blades; the number of blades of the first compressor stage is less than the number of blades of the second compressor stage; there is an annular gap between the first row of blades and the second row of blades; the first compression stage is designed so to assure that the CO2 flow is in supercritical condition, preferably close to CO2 critical point, at its outlet, and so that the second compressor stage process CO2 in supercritical condition.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. Ser. No. 18/253,638, filed on May 19, 2023, and entitled “COMPRESSOR FOR CO₂ CYCLE WITH AT LEAST TWO CASCADE COMPRESSIONS STAGES FOR ASSURING SUPERCRITICAL CONDITIONS,” which is a national stage application of PCT/EP2021/025459, filed on Nov. 24, 2021, which claims priority to Italian Patent Application No. 102020000028685, filed on Nov. 27, 2020. The content of these applications is incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The subject-matter disclosed herein relates to a CO₂ flow compressor, an CO₂ cycle energy generation system and a method for compressing a CO₂ flow.

BACKGROUND

[0003] The European Union, in short EU, has set a long-term goal to cut greenhouse gas emissions by 80-95% by 2050 compared to 1990 levels. The EU 2050 Energy Strategy has therefore serious implications for our energy system and includes new challenges and opportunities. This is a general trend all over the world.

[0004] Renewable energies (such as wind and solar) are moving to the center of the energy mix in Europe and raise the question of grid stability in the event of large power output fluctuations. In this context, enhancing the flexibility and the performance of conventional power plants is seen as a good opportunity to both secure the energy grid while reducing their environmental impact.

[0005] The sCO₂-flex consortium, made up of 10 experienced key players from 5 different EU member states, seeks to increase in the operational flexibility (fast load changes, fast start-ups and shutdowns) and efficiency of existing and future coal and lignite power plants, thus reducing their environmental impacts, in line with EU targets.

[0006] Supercritical carbon dioxide (sCO₂) is a fluid state of carbon dioxide where it is held at or above its critical temperature and critical pressure. The fluid presents interesting properties that promise substantial improvements in conventional power plant system efficiency.

[0007] sCO₂ based technology has the potential to meet EU objectives for highly flexible and efficient conventional power plants, while reducing greenhouse gas emissions, residue disposal and also percentage water consumption reduction.

[0008] A sCO₂ cycle is a closed cycle wherein the fluid is compressed by one or more compressors, heat is introduced into the cycle by a first heat exchanger, the fluid is expanded by one or more expanders and heat is released to the environment through a second heat exchanger. Advantageously, after the expansion and before release of heat to the environment, the fluid passes through a third heat exchanger, i.e. a recovery heat exchanger, to improve the efficiency of the cycle.

SUMMARY

[0009] Usually, the first compressor of sCO₂ cycle works with a CO₂ flow close to critical point. Then, the sCO₂ cycle presents reduced work of CO₂ compressor, that take advantages from the

real gas behavior of the working fluid near the critical point. This feature enhances increasing the overall thermal efficiency of the sCO₂ cycle. However, there is large variation of CO₂ properties very close to critical point, having technological implications on the design of turbomachinery and heat exchangers.

[0010] In particular, the CO₂ flow reaches the impeller of the first compressor in a multiphase status because of local acceleration upstream and across compressor impeller leading edge, due to the size of the blade channels of the impeller. In multiphase region, i.e. under the saturation dome, the speed of sound steeply decreases causing the creation of a sonic region with consequent limitation of compressor operating range.

[0011] When a fluid flowing at a given pressure and temperature passes through a constriction, the fluid velocity increases. At the same time, the Venturi effect causes the static pressure, and therefore the density, to decrease at the constriction. This may result in the creation of a sonic region, which limits the compressor operating range.

[0012] This problem is heightened in presence of numerous compressor blades, i.e. in presence of numerous constrictions at the compressor inlet due to blade channels.

[0013] Due to Venturi effect, a high number of blades at the inlet of the stage increase local flow acceleration that, combined with large departures from the ideal-gas behavior approaching the critical point, could promote phase-change phenomena of the CO₂, reducing compressor efficiency and cycle efficiency.

[0014] According to one aspect, the subject-matter disclosed herein relates to a compressor arranged to process a CO₂ flow, comprising a first compressor stage and a second compressor stage, downstream the first compressor stage; the first compressor stage comprises a first row of rotary blades with a first number of blades and the second compressor stage comprises a second row of rotary blades with a second number of blades; the first number of blades is less than the second number of blades; the CO₂ flow is in supercritical condition at the outlet of the first compressor stage.

[0015] In particular, the trailing edge of the blades of the first stage discharges a CO₂ flow directly to an annular gap and the leading edge of the blades of the second stage receives a CO₂ flow directly from the annular gap, CO₂ pressure at the first compressor stage trailing edge being equal or higher than saturation pressure plus a predetermined pressure margin, said pressure margin being related to pressure drop inside the second compressor stage.

[0016] According to another aspect, the subject-matter disclosed herein relates to an energy generation system based on a supercritical CO₂ cycle and including a compressor with at least two cascade compression stages for assuring supercritical conditions, and an annular gap in-between.

[0017] According to still another aspect, the subject-matter disclosed herein relates to a method for compressing CO₂ flow; a first compression step is used for compressing said CO₂ flow to a supercritical condition through a first compressor stage (200) so to generate a supercritical CO₂ flow, and a second compression step is used for compressing said supercritical CO₂ flow through a second compressor stage (300); the first compression step is such that, at the end of compression, CO₂ is close to critical point; between the first compression step and the second compression step there is an isoenthalpie step that maintains substantially constant both total pressure and static pressure, in other words: low losses for total pressure and recovery for static pressure

Description

DRAWINGS

[0018] A more complete appreciation of the disclosed embodiments of the invention and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the

accompanying drawings, wherein:

[0019] FIG. 1 shows a schematic view of a CO₂ system,

[0020] FIG. 2A shows a perspective view of the compressor of FIG. 1,

[0021] FIG. 2B shows a lateral view of the compressor of FIG. 1,

[0022] FIG. 3 shows an enlarged view of a portion of FIG. 2A,

[0023] FIG. 4 shows cross-sectional schematic view of a compression system for a CO₂ flow cycle, and

[0024] FIG. 5 shows an example of a CO₂ compression on a T-s diagram.

[0025] In the various figures, the same elements will be indicated with the same reference number.

DESCRIPTION

[0026] The subject matter herein disclosed relates to a compressor and a CO₂ system working with a CO₂ flow, a method for compressing CO₂ flow and a compressor assembly for a CO₂ flow cycle.

[0027] The efficiency of a gas turbine cycles mainly depends on its pressure ratio (i.e. the ratio between the pressure of the gas flow at the compressor inlet and at the compressor outlet). The maximum pressure is limited due to the cost related to the piping and measurement systems; thereby the minimum pressure of the sCO₂ cycle significantly influences the cycle efficiency.

[0028] At the same time, the efficiency of the cycle is affected also by the condition of the gas flow, in particular at the inlet of the compressor. In fact, fixed the maximum cycle pressure due to costs, working close to the critical point is advantageous because it allows the compression work to decrease, having as results the improving of cycle efficiency.

[0029] However, in a CO₂ condition near to critical point, shock waves may occur, limiting the operating region of the compressor and decreasing the efficiency.

[0030] In order to overcome this, the compression system disclosed hereby aims to increase cycle efficiency by increasing the pressure of the fluid just enough to allow the compressor impeller to work away from the critical point while maintaining a high pressure ratio.

[0031] This is accomplished by having an inducer stage that compresses the fluid with a small pressure ratio and that is designed with a low number of blades to limit the problem of shock waves that causes the collapse of performance. Advantageously, having an inducer stage that has low number of blades avoids the compressor inlet to become the sonic throat of the component.

[0032] Reference now will be made in detail to embodiments of the disclosure, an example of which is illustrated in the drawings.

[0033] The example is provided by way of explanation of the disclosure, not limitation of the disclosure. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure without departing from the scope or spirit of the disclosure.

[0034] According to one aspect and with reference to FIG. 1, the subject-matter disclosed herein provides an energy generation system based on a supercritical CO₂ cycle, i.e. a gas turbine plant working with CO₂ as a working fluid mainly in supercritical conditions. Typically, in this type of cycle the working fluid at minimum cycle pressure is in supercritical conditions, but also working fluid in subcritical multiphase conditions with minimum cycle pressure between 80-100% of critical pressure are allowed.

[0035] The CO₂ system of FIG. 1 comprises two heat exchangers **2000A**, **2000B**, an expander **3000** and a compressor **1000**; advantageously, the compressor and the turbine are driven on the same shaft **1010**. The shaft **1010** determines an axis A corresponding to the main development direction of the shaft **1010**. As used herein, the terms “axial” and “radial” refers respectively to a direction parallel and perpendicular to the axis A.

[0036] Referring to FIG. 1, the CO₂ flow flows in a clock-wise direction: is compressed by a compressor **1000**, is heated in a first heat exchanger **2000A**, is expanded by an expander **3000**, is cooled in a second heat exchanger **2000B** and finally restarts the cycle. In other words, the CO₂ system is a closed-cycle gas turbine.

[0037] According to a preferred embodiment, the CO₂ system comprises a third heat exchanger **2000C**, called also “recuperator”; the third heat exchanger **2000C** is suitable for increasing the thermal efficiency of the cycle, receiving the CO₂ flow at the outlet of compressor **1000** as a cold fluid and the CO₂ flow at the outlet of the expander **3000** as a hot fluid. The recuperator **2000C** allows to recover waste heat from expander exhaust CO₂ flow and use it to pre-heat the compressed CO₂ flow from the compressor **1000** before further heating of compressed CO₂ flow in the heat exchanger **2000A**, reducing the external heat required.

[0038] In the example of FIG. 1, the expander **3000**, in particular the shaft **1010** that drives the expander **3000**, is coupled with an electric generator **4000**, in particular an alternator; alternatively, the expander **3000** may be connected to an external load not shown in figure.

[0039] It is to be noted that, depending on the design of the cycle, the number of the machines and heat exchangers may vary, as well as the number of shafts driving the machines.

[0040] According to one aspect and with reference to FIGS. 2 and 3, the subject-matter disclosed herein provides a compressor **1000** to be used for example in a supercritical CO₂ system for generating electric energy or for supplying an external load.

[0041] The compressor **1000** comprises a first compressor stage **200** and at least a second compressor stage **300** downstream the first compressor stage **200**. It is noted that “stage” is here referred to a single row of blades, which can be stationary or rotary. For example, if there is a first row of rotary blades and a second row of stationary blades, the first row of rotary blades is a first stage and the second row of stationary blades is a second stage.

[0042] The first compressor stage **200** comprises a first row of rotary blades **250**; the second compressor stage **300** comprises a second row of rotary blades **350**. Preferably, the first row of blades **250** has an inducer type of blades and the second row of blades **350** has an exducer type of blades. In a preferred embodiment shown in FIGS. 2 and 3, the first number of blades is less than the second number of blades.

[0043] Preferably, the first number of blades is about one-half or about one-third the second number of blades. For example, if the second row of blades **350** has a number of blades equal to **18**, the number of blades of the first row of blades **250** may be for example 11 or 10 or 9 or 8 or 7 or 6. It is to be noted that the ratio between these two numbers may be any number typically different from an integer number; for example, it may be higher than 1 and lower than 2 or higher than 2 and lower than 3. Therefore, the numbers of blades may be freely chosen independently depending on the mechanical design and performance desired for the two compression stages.

[0044] The compressor **100** works typically with a CO₂ flow and the first compressor stage **200** provides at the outlet a CO₂ flow in supercritical conditions, wherein with “supercritical conditions fluid” is defined a fluid having pressure above its critical point, i.e. having pressure higher than its critical pressure.

[0045] In other word, at the outlet of the first compressor stage **200**, the CO₂ flow has pressure higher than about 7.37 MPa.

[0046] Specifically and with reference to FIG. 3, the first compressor stage **200** is arranged to provide a pressure increase between a leading edge **210** with a bottom **212** and a top **214** and a trailing edge **220** with a bottom **222** and a top **224** of the first row of blades **250**; such pressure increase being enough for the CO₂ flow at the trailing edge **220** to reach supercritical conditions.

[0047] Preferably, the CO₂ flow has a higher pressure at the trailing edge **220** with respect to the pressure at the leading edge **210**. The ratio between the outlet pressure and the inlet pressure of a flow passing through a compressor stage is known as “pressure ratio” or “compression ratio”.

[0048] Preferably, the leading edge **210** of the first row of blades **250** is in correspondence of an inlet section of the compressor **1000**, said inlet section receiving a suction CO₂ flow. The CO₂ flow is then discharged in correspondence of the trailing edge **220** of the first row of blades **250**.

[0049] Preferably, the first row of blades **250** has mainly axial development with respect to a direction determined by axis A. Specifically, the axial development of first row of blades **250** is

such that the CO₂ flow flows mainly in axial direction.

[0050] With reference to FIGS. 2 and 3, the compressor **1000** comprises a second compressor stage **300** downstream the first compressor stage **200**. Specifically, the second compressor stage **300** is arranged to provide a pressure increase between a leading edge **310** and a trailing edge **320** of the second row of blades **350**, such pressure increase being much higher than the pressure increase provided between the leading edge **210** and the trailing edge **220** of the first compressor stage **200**.

[0051] In other words, the pressure ratio of the first compressor stage **200** is much smaller than the pressure ratio of the second compressor stage **300**, i.e. the second compressor stage **300** provides the main pressure ratio of the overall pressure ratio of CO₂ cycle. Preferably, the pressure ratio of the first compressor stage **200** is less than 70% of the pressure ratio of the second compressor stage **300** and possibly is more than 3% of the pressure ratio of the second compressor stage **300**; for example, the first pressure ratio may be equal to approximately 1.1 and the second pressure ratio may be equal to approximately 1.7.

[0052] In a preferred embodiment and with reference to FIGS. 2, 3 and 4, the second compressor stage **300** is a centrifugal compressor stage, having both axial and radial development with respect to a direction determined by axis A. In particular, the flow-path between the leading edge **310** and the trailing edge **320** defines a substantially twisted surface with respect to a direction determined by axis A. Specifically, the leading edge **310** and the trailing edge **320** are located at a different radial distance from axis A.

[0053] The first compressor stage **200** (in particular the first row of blades **250**) is arranged to provide the CO₂ flow directly to the second compressor stage **300** (in particular to the second row of blades **350**) without any stationary component in-between, in particular any stator blade, passing through an axial annular gap **400**. Specifically, the CO₂ flow flows from the first row of blades **250** to the second row of blades **350** without any (substantial) change in pressure (both in static pressure and in total pressure), for example due to stator blades between the trailing edge **220** and the leading edge **310**. The Applicant has realized that stator blades between two consecutive rows of rotor blades, which is very common in turbomachines, might seem beneficial; however, in the present case, in order to avoid a system throat, “blade solidity” should be low and the benefit on static pressure recovery would be negligible.

[0054] The second row of blades **350** is axially spaced from said first row of blades **250**. Specifically, an axial annular gap **400** (developing around axis A) is located between the trailing edge **220** of the first row of blades **250** and the leading edge **310** of the second row of blades **350**. In this way, wakes relax so that strong aeromechanic interaction between the two rows is avoided.

[0055] Preferably, the axial annular gap **400** between the trailing edge **220** and the leading edge **310** has a length between one and two times the height of trailing edge **220** of the first row of blades **250**.

[0056] With reference to FIGS. 2 and 3, the trailing edge **220** of the first row of blades **250** and the leading edge **310** of the second row of blades **350** may be not aligned along an axial direction. In particular, the leading edge **310** of the second row of blades **350** may have different circumferential positions with respect to the trailing edge **220** of the first row of blades **250** (this arrangement is known as “clocking effect”).

[0057] In a preferred embodiment, the compressor **1000** comprises a rotor, the first row of blades **250** and the second row of blades **350** being part of the rotor.

[0058] With reference to FIG. 4, the rotor is preferably driven by the shaft **1010**, so that the first row of blades **250** and the second row of blades **350** rotates at the same angular velocity.

[0059] In an alternative embodiment, the compressor **1000** comprises a first rotor and a second rotor, the first row of blades **250** being part of the first rotor and the second row of blades **350** being part of the second rotor.

[0060] Advantageously, the first rotor is driven by a first shaft and the second rotor is driven by a second shaft, the first shaft and the second shaft rotating at different angular velocity.

[0061] With reference to FIG. 4, the compressor **1000** may further comprise inlet guide vanes **100** upstream the first row of blades **250**. Advantageously, the inlet guide vanes **100** comprise a stator row of blades; the stator row of blades can be fixed or can vary blades angle of attack, regulating the CO₂ flow sucked by compressor **1000**.

[0062] According to another aspect, the subject-matter disclosed herein relates to a method for compressing CO₂ flow using a compressor for example similar or identical to compressor **1000** described above; such method may be implemented in an energy generation system based on a supercritical CO₂ cycle similar or identical to the energy generation system described above.

[0063] The method comprises an initial step of compressing CO₂ flow to supercritical conditions through a first compressor stage **200** and a following step of compressing supercritical CO₂ flow through at least a second compressor stage **300**; between the first compression step and the second compression step there is a low-loss isoenthalpic step (in particular inside the axial annular gap **400**) that maintains substantially constant both total pressure and static pressure.

[0064] The initial step of compressing CO₂ flow to supercritical conditions is such that, at the end of compression, the thermodynamic state point of CO₂, on a T-s diagram or equivalent, is located outside the saturation dome, approximately near the CO₂ critical point (P_c, T_c).

[0065] With reference to FIG. 5, it is shown a CO₂ temperature-entropy diagram wherein is highlighted the CO₂ critical point (P_c, T_c) as a black point at the top of the saturation dome. According to the method disclosed herein, after the initial step of compressing CO₂ flow, the thermodynamic state point of CO₂, i.e. the point which represent the thermodynamic state of the CO₂ defined by at least two state variables (for example temperature and pressure), is located outside the saturation dome, in particular around the highlighted area **800** above the CO₂ critical point (P_c, T_c).

[0066] In a preferred embodiment, the pressure at the outlet of the first compressor stage **200** is equal or higher than saturation pressure plus a predetermined pressure margin, said pressure margin being related to pressure drop inside the second compressor stage **300**.

[0067] It has to be noted that the initial step of compressing CO₂ flow to supercritical conditions can be followed by one or more following steps of compressing supercritical CO₂ flow; preferably, the initial step of compressing CO₂ flow has a pressure ratio much smaller than each following step.

[0068] According to another aspect, the subject-matter disclosed herein relates to compressor arranged to process a CO₂ flow comprising: [0069] a first rotating compressor stage comprising a first row of inducer blades, said inducer blades extending mainly axially, having a leading edge (**210**) and a trailing edge (**220**); [0070] a second rotating compressor stage comprising a second row of exducer blades extending mainly axially or mainly radially or both axially and radially, having a leading edge (**310**) and a trailing edge (**320**); [0071] an axial annular gap **400** between the first rotating compressor stage and the second rotating compressor stage.

[0072] In a preferred embodiment, the inducer trailing edge (**220**) discharge a CO₂ flow directly to the axial annular gap **400** and the exducer leading edge (**310**) receive a CO₂ flow directly from the axial annular gap **400**. Preferably, the CO₂ flow pressure at the inducer trailing edge (**220**) is higher than the CO₂ flow pressure at the inducer leading edge (**210**). In particular, the CO₂ flow pressure at the trailing edge (**220**) is equal or higher than saturation pressure plus a predetermined pressure margin, said pressure margin being related to pressure drop inside the second rotary compressor stage.

[0073] The above-mentioned pressure margin aims at avoiding that saturation conditions are reached by the CO₂ flow inside the second compressor stage. Theoretically, there is no pressure drop within a compressor stage. However, in practice, there may be some pressure drop shortly after the leading edge (**310**) of the second row of exducer blades; the regions mostly at risk from this point of view are on suction side of exducer blades, near the leading edge (**310**).

[0074] The minimum pressure value inside the second row of exducer blades strongly depends on

design choices and typically is between 90% and 50% of the total inlet pressure at the second rotating compressor stage, i.e. at the leading edge (310).

[0075] While aspects of the invention have been described in terms of various specific embodiments, it will be apparent to those of ordinary skill in the art that many modifications, changes, and omissions are possible without departing from the spirit and scope of the claims. In addition, unless specified otherwise herein, the order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments.

Claims

1. A method for compressing a CO₂ flow using a compressor, comprising: a first compression step for compressing the CO₂ flow to a supercritical condition through a first compressor stage so to generate a supercritical CO₂ flow; a second compression step for compressing the supercritical CO₂ flow through a second compressor stage; wherein the first compression step is such that, at the end of compression, CO₂ is close to critical point (P_c , T_c); wherein between the first compression step and the second compression step there is an isoenthalpic step that maintains substantially constant both total pressure and static pressure.
2. The method of claim 1, wherein the first compression step is such that, at the end of compression, thermodynamic state point of CO₂, on a T-s diagram, is located outside the saturation dome, approximately near CO₂ critical point (P_c , T_c).
3. The method of claim 2, wherein at the end of the first compression step, the pressure is equal or higher than saturation pressure plus a predetermined pressure margin, the pressure margin being related to pressure drop inside the second rotary compressor stage.
4. The method of claim 1, wherein the first compression step is followed by one or more compressing steps of compressing CO₂ flow.
5. The method of claim 1, wherein the first compression step has a pressure ratio smaller than the second compression step.
6. A method of compressing a fluid, comprising: in a compressor, providing a rotor having a first row of blades and a second row of blades spaced axially apart from the first set of blades to create a gap; directing the fluid through the first row of blades; directing the fluid from the first row of blades across the gap; and directing the fluid from the gap through the second row of blades after the gap.
7. The method of claim 6, wherein the first row of blades is at a different circumferential position than the second row of blades.
9. The method of claim 6, wherein the trailing edge of the first row of blades is circumferentially offset from the leading edge of the second row of blades.
10. The method of claim 6, wherein the trailing edge of the first row of blades and the leading edge of the second row of blades are not axially aligned.
11. The method of claim 6, wherein the gap resides between the trailing edge of the first row of blades and the trailing edge of the second row of blades.
12. The method of claim 6, wherein the gap has a length that is at least equal to a height of a leading edge of the first row of blades.
13. The method of claim 6, wherein the gap has a length that is less than two times a height of a leading edge of the first row of blades.
14. The method of claim 6, wherein the rotor has twice as many blades in the second row as the first row.
15. A method, comprising: compressing a fluid by, directing a flow of fluid across an inducer type blade found on a rotor in a compressor; and then, directing the flow of fluid across an exducer type of blade found on the rotor downstream of the inducer type of blade, wherein the flow of fluid transits a gap after exiting the inducer type of blade and before entering the exducer type of blade.

- 16.** The method of claim 15, wherein the gap has a length that at least equal to a height of a leading edge of the inducer type of blade.
- 17.** The method of claim 15, wherein the gap has a length that is less than two times a height of a leading edge of the inducer type of blade.
- 18.** The method of claim 15, wherein the inducer type of blade and the exducer type of blade are circumferentially offset from one another.
- 19.** The method of claim 15, wherein the inducer type of blade has a trailing edge that is circumferentially offset from a leading edge of the exducer type of blade.
- 20.** The method of claim 15, further comprising: directing the flow of fluid through inlet guide vanes found upstream of the inducer type of blades.
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